

Enzo Miguel Toledo Venegas

Mathematical Engineering and M.Sc. in Data Science, University of Chile

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Teamwork

Mathematical Modeling

Adaptability

Professional Summary

Mathematical Engineering and Data Science M.Sc. student at the University of Chile, within a joint-degree program, with an interest in the rigorous study of the mathematical foundations of machine learning, oriented toward the formalization of methods that currently operate on heuristic grounds. I have a strong theoretical background in analysis, probability, and mathematical optimization, together with practical experience in probabilistic modeling, simulation, optimization, and the development of machine learning solutions in both academic and industry settings.

Advanced coursework: Deep Generative Models, Machine Learning, Natural Language Processing, Optimal Control, Mixed-Integer Linear Programming, Markov Processes, Stochastic Calculus, Stochastic Simulation, Numerical Analysis of Partial Differential Equations.

Advanced Technical Skills

- **Languages:** English (professional working proficiency), Mandarin Chinese (conversational).
- **Programming:** Python, R, Julia, SQL (PostgreSQL). **Tools:** PyTorch, scikit-learn, JuMP, Git, PowerBI

Work Experience and Extracurricular Activities

- **External Researcher, FollowUp Customer Experience** (March – July 2026) — Design and implementation from scratch of a computer vision pipeline for stockout alerts. Modular architecture based on YOLO and Depth Anything V2, integrating homographic rectification, planograms as priors, product detection and classification, and geometric comparison against a reference layout to trigger alerts.
- **Machine Learning Intern, In-Data** (January – February 2026) — Evaluation and comparison of open-source multimodal models for structured data extraction from building permits under local hardware constraints.
- **Financial Risk Analyst, Scotiabank** (January – March 2025) — Analysis, monitoring, and automation of tracking processes for the bank's provisioning and management models.
- **Eolian Project (Solar Car), University of Chile** (August – December 2024) — Analysis of electric motor data for its modeling and optimization, within the motors and energy models areas.

Selected University Projects

- **Deep Learning as an Optimal Control Problem** (Julia) — Formalization of ResNet training as a continuous-time optimal control problem.
- **Approximation of SPDEs via Wiener–Chaos and Fourier Neural Operators** (Python) — Framework combining Wiener–Chaos expansions with Fourier Neural Operators, decomposing the stochastic solution into coupled deterministic coefficients. Cases: bistable heat equation with stochastic forcing and 2D stochastic Navier–Stokes.
- **Estimation of Critical Percolation Parameters** (Python) — Bernoulli percolation on square grids via Monte Carlo, estimating the critical threshold and the critical exponent near the phase transition, using the Restart technique for low-probability events.